

$$Y_{it} = \beta_0 - 0.50Y_{0i} + 0.30X_{1i} - 0.30X_{2i} + 0.40B_{1i} + 0.30B_{2i} + 0.25B_{3i} + 0.35B_{4i} + 0.05t + 0.70A_i - C_i b_t(Z_i) + \varepsilon_{it}, \quad t = 1, \dots, 12,$$

$$\boldsymbol{\varepsilon}_i \sim N_{12}(\mathbf{0}, \Sigma), \quad \Sigma = 4\{(1 - 0.30)I_{12} + 0.30\mathbf{1}\mathbf{1}^\top\}.$$